

ROV Simulation Validation Report

Topic: Dynamic Modelling of a Sensor Integrated Remotely Operated Underwater Vehicle for Environment Modelling and Mapping of Seabed

By: H035, H040, H046, H047

1. Introduction

This report outlines the validation of an underwater Remotely Operated Vehicle (ROV) simulation created with PyBullet. The goal is to ensure that the physics in the simulation reflect real underwater dynamics accurately.

The validation process consists of three main components: comparison with standard textbook equations, testing energy consistency, and conducting a sensitivity analysis on key physical parameters—like fluid density and drag coefficient.

Underwater vehicle dynamics are complex due to the interplay of fluid forces and rigid body motion. This simulation aims to capture those interactions by using simplified but meaningful models.

2. Textbook Comparison

The simulation includes buoyancy, drag, thrust, and restoring moments, which are compared with standard equations from fluid mechanics and marine engineering.

Buoyancy Force

- Simulation:

$$F_b = \rho V g \lambda$$

- Textbook:

$$F_b = \rho V g$$

The multiplier λ accounts for real-world variations such as imperfect geometry and density changes.

Drag Force

- Simulation:

$$F_d = -\frac{1}{2} \rho C_d A v |v|$$

- Textbook:

$$F_d = \frac{1}{2} \rho C_d A v^2$$

The negative sign ensures drag always opposes motion.

Rotational Drag

$$\tau_d \propto \omega |\omega|$$

This quadratic dependence matches fluid resistance behavior.

Restoring Moment

$$\tau = r \times F_b$$

where r is the distance between the center of buoyancy and center of mass.

Overall, the simulation equations closely match established theoretical models.

3. Energy Validation

Energy validation ensures consistency with conservation principles.

Kinetic Energy

$$T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Potential Energy

$$V = mgz$$

Total Energy

$$E = T + V$$

Thruster Power

$$P = \mathbf{F} \cdot \mathbf{v}$$

Drag Loss

Energy dissipated due to drag:

$$E_{\text{loss}} = \int F_d \cdot v \, dt$$

The total work done by thrusters satisfies:

$$E_{\text{input}} \approx E_{\text{stored}} + E_{\text{loss}}$$

A small residual error confirms numerical stability.

Drag Loss:

Energy is lost through fluid resistance. The total work done by the thrusters is compared with total energy and drag losses. A small residual difference indicates numerical stability and correctness of implementation. This validation demonstrates that the system adheres to expected physical laws.

4. Sensitivity Analysis

Sensitivity analysis examines how tweaks in parameters impact system performance.

Fluid Density:

Raising density enhances buoyancy, letting the ROV rise more easily.

Drag Coefficient:

Increased drag slows speed and heightens energy loss.

Buoyancy Multiplier:

A higher multiplier boosts upward force and stabilizes the ROV.

These results show the simulation reacts realistically to parameter changes.

Sensitivity analysis is key to ensuring the model is robust and dependable.

5. Discussion

The simulation successfully captures essential aspects of underwater motion. Using quadratic drag models and distributed thruster forces provides a realistic depiction of vehicle behavior.

The restoring moment is vital for maintaining stability, especially during disturbances.

However, limitations exist, including simplified geometry and a lack of complex fluid phenomena like turbulence and vortex shedding.

Despite these limitations, the model establishes a solid foundation for future development and testing.

6. Conclusion

The validation of ROV simulation was performed based on comparing with analytical expressions, energy verification, and sensitivity analysis.

It has been shown that the simulated system works realistically from a physical point of view.

Thus, the model can be successfully used in education and experiments concerning underwater robotics.